Lecture 37 — Why did the system behave that way?

Fri Nov 20

Last updated: October 29, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Strang / MIT (optional): Reading: Strang §6.1 (PDF) — eigenvalues
Video: 18.06 Lec 21 — eigenvalues and eigenvectors · MIT 18.06 playlist (Strang)

Reading: Strang §6.1 and Lachniet Ch. 5 (eigenvalues/eigenvectors). Related objectives: Lectures. Schedule.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**. Same day: Lab 5 closes the Lab 4 campus mystery with $\lambda = 1$.

In the story. We can name eigenpairs (Mon) and find them by hand (Wed). Today: **software**, **dominant** $|\lambda|$ explains long-run $A^t x_0$, and Lab 5 matches campus x_{20} to the $\lambda = 1$ direction.

Learning objectives

By the end of class, you should be able to:

1. Compute eigenvalues and eigenvectors of a general matrix with software.
2. Verify $Av \approx \lambda v$ numerically.
3. Use dominant eigenbehavior to explain a simulated system (including Lab 4).
4. Interpret complex eigenvalues as rotation/oscillation behavior.
5. Explore a symmetric-matrix or normal-mode example and observe orthogonal eigenvectors.

Warm-up

Let

$$A = \begin{bmatrix} 1.1 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}, \qquad v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Hand-compute Av. Guess a plausible λ so that $Av \approx \lambda v$. (Exact eigenpairs come from software in lab.)

Fill in

Software eigenpairs. Octave `eig(A)` / Python `np.linalg.eig(A)` return eigenvalues and a matrix whose columns are eigenvectors.

Verify. For each reported pair (λ, v), check

$$Av \approx _____$$

(componentwise; floating-point noise is expected). Never trust a bare printout without this check.

Dominant behavior. If $|\lambda_1| > |\lambda_2| \geq \cdots$ and x_0 has a component along the λ_1-eigenvector, then for large t

$$A^t x_0 \approx c\,\lambda_1^t\, v_1.$$

That explains Lab 4-style growth/decay/stabilization once you know the eigenpairs.

Complex eigenvalues. For a real matrix, nonreal eigenvalues come in conjugate pairs. In the real plane they produce _____ (rotation / oscillation), matching Monday's pure-rotation example.

Normal modes (demo, not a DE course). For a symmetric mass–spring style matrix, eigenvectors are often _____ (orthogonal). Software shows the modes; we do **not** derive differential equations today.

Worked in class

W1. Software sketch. On the board / laptop, for $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$:

```
# Octave:  [V,D] = eig(A)     or    eig(A)
# Python:  vals, vecs = np.linalg.eig(A)
```

Read off λ's and columns of V. Verify $Av_1 = \lambda_1 v_1$ for one column.

W2. Explain a simulation. Same A as W1; $x_0 = (1, 0)$. Using eigenpairs, explain in one sentence why x_t grows and which direction it approaches.

W3. Complex / oscillation. Software on $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ should report eigenvalues near $\pm i$. Connect that to the

cycling you saw in Lab 4 stretch / Lecture 35.

W4. Symmetric demo. Let $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ (symmetric). Check that the two eigenvectors from W1 are orthogonal ($v_1^{\mathsf{T}} v_2 = 0$). Normal-mode pictures use the same idea in higher dimension.

Your turn

Y1. Pseudocode: given A, obtain eigenpairs in software and write the one-line check that $Av \approx \lambda v$ for the first pair.

Y2. True or false: if $|\lambda_1| = 1.5$ and $|\lambda_2| = 0.5$, then for generic x_0 the long-run direction of $A^t x_0$ is the λ_1-eigendirection.

Answers (Your turn)

Y1. Sample: `vals, V = np.linalg.eig(A)` then `A @ V[:,0] - vals[0]*V[:,0]` (should be near 0); Octave: `[V,D]=eig(A)` then `A*V(:,1) - D(1,1)*V(:,1)`.

Y2. **True.** The λ_1 term grows relative to λ_2, so the state aligns with v_1 (unless the v_1-component of x_0 is exactly zero).